Mahmoud Yosri

Python Developer

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PROFILE

Python developer who builds complete products end to end: **FastAPI** back ends on **PostgreSQL**, data pipelines, and the machine learning inside them. I write code that is tested against real data, runs on modest hardware, and is simple for one person to operate: one script to set up and start, clear logs, and safe defaults.

PROJECTS

Studio: Offline AI Product Photo Pipeline | Code | Case study *2026*

- Built an open-source (MIT) tool that removes backgrounds and matches product photos fully offline: **ONNX Runtime** on the CPU with IS-Net and U^2-Net-P, **NumPy** and **Pillow**.
- Split the work into a slow AI step that runs once (**1.2 s**) and a fast render step on every edit (**0.4 s**), with the mask cached by a small **FastAPI** service.
- Wrote a known-truth test bench on 8 real photos: dark and blurry photos flagged **8 of 8**; one-tap white balance cuts color error (ΔE) from **10.6 to 3.5**.

inPace: Learning Accountability App | Code | Case study *2026*

- Shipped **v1.14** of an open-source (MIT) app for **Windows, Mac, Linux and Android**: a local **Starlette** server in its own app window, **Jinja** pages, and an append-only **SQLite** history.
- Made progress tamper-evident: milestones are signed with **Ed25519 (PyNaCl)** and chained; checkers verify through links and files, and a signed course library publishes through bot-checked pull requests.
- Set up **GitHub Actions** to run **101 tests** on three systems, build all four apps (including a signed Android APK), and publish **SHA-256** sums the auto-updater checks.

Argon: Job-Search Assistant and Market Analysis | Case study *2026*

- Built a system that collects job ads from Telegram channels, WhatsApp groups and job sites (**4,834 jobs**) and scores each ad from 0 to 100 against the user's CVs.
- Analysed **3,824 ads over 90 days** to rank in-demand skills, with a local LLM (**Ollama**) and **AES-256** encrypted backups that restore into SQLite or PostgreSQL.

Rubies: Multi-Shop Marketplace API | Case study *2026*

- Designed the **FastAPI**, **SQLAlchemy 2** and **PostgreSQL 17** back end for four roles (customer, shop, courier, admin) with **Alembic** migrations.
- Implemented QR and 6-digit PIN delivery handover, partial returns, cash settlement, and sign-in codes over Telegram and WhatsApp instead of SMS.
- Covered the system with **24 pytest** API tests and **14 Playwright** browser tests, plus an engineering handbook with Mermaid diagrams.

Hydro Alex: Online Store for a Plumbing Shop | Case study *2026*

- Enforced order steps on the server so each status can only move to allowed next steps, and prevented double-selling with a single conditional stock update.
- Built money reports with PDF and Excel export, and a **Docker Compose** and one-script setup; tested with **20 pytest** tests on a real PostgreSQL database.

TECHNICAL SKILLS

- **Python & Back End:** Python, FastAPI, Starlette, Uvicorn, Jinja, SQLAlchemy 2, Alembic, REST APIs, CLI tools.
- **Data & Databases:** PostgreSQL, SQLite, SQL, Pandas, NumPy, Pillow.
- **ML:** ONNX Runtime, PyTorch, TensorFlow, Scikit-Learn, XGBoost, Ollama.
- **Testing & Tools:** pytest, Playwright, GitHub Actions, Docker Compose, Bash, Git, PyNaCl (Ed25519).
- **Other Languages:** C++, C#, SQL, Dart, JavaScript, TypeScript (Next.js).
- **Spoken Languages:** Arabic (Native), English (B2).

EDUCATION

Bachelor of Computer Science Sep 2020 – Jun 2024
Kafrelshikh University, Faculty of Computer Science and Information Technology **GPA:** 2.9/4.0

CERTIFICATIONS

- **Python for Data Analysis** | NTI Verification *2024*
- **Machine Learning Specialization** | Coursera Verification *2024*
- **Deep Learning Specialization** | Coursera *2025 – 2026*